Vadim Kolotenko
Haaga-Helia ammattikorkeakoulu
Linux-palvelimet - ICI003AS2A-3021

Raportti 5
9.10.2026

Intro

site name is linuxcourse.pitzuna.com. Apache still listens on port 3456, and Nginx Proxy Manager still owns ports 80 and 443.

DNS

pitzuna.com is served through Cloudflare, so a new name was required. I added an A record for linuxcourse and set it to DNS only. dig from the Debian virtual machine then returned 84.248.108.122, both from the local resolver 1.1.1.1 and from 8.8.8.8.

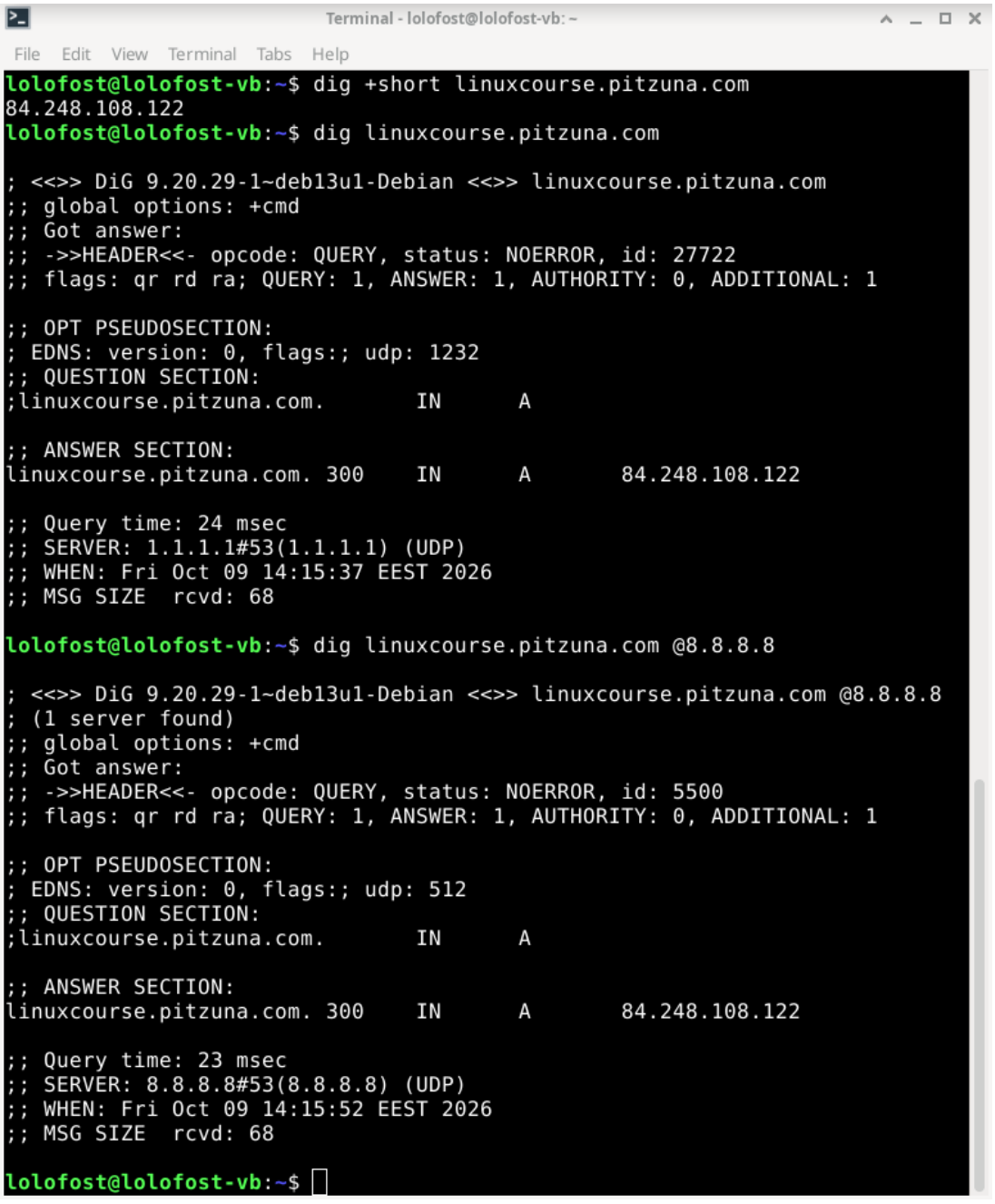


Image 1. The name points at the home address.

I captured DNS on eno1 while the lookups ran. Both queries left the server toward 8.8.8.8 port 53, and both answers were 84.248.108.122.

```
pitzuna@zunaserver:~$ sudo tcpdump -i eno1 port 53 -n -v
tcpdump: listening on eno1, link-type EN10MB (Ethernet), snapshot length 26
2144 bytes
14:16:59.503028 IP (tos 0x0, ttl 64, id 3880, offset 0, flags [none], proto
 UDP (17), length 80)
    192.168.0.100.58874 > 8.8.8.8.53: 49366+ [1au] A? linuxcourse.pitzuna.c
om. (52)
14:16:59.529876 IP (tos 0x0, ttl 122, id 26770, offset 0, flags [none], pro
to UDP (17), length 96)
    8.8.8.8.53 > 192.168.0.100.58874: 49366 1/0/1 linuxcourse.pitzuna.com.
A 84.248.108.122 (68)
14:17:17.505656 IP (tos 0x0, ttl 64, id 48932, offset 0, flags [none], prot
o UDP (17), length 92)
    192.168.0.100.57202 > 8.8.8.8.53: 61419+ [1au] A? linuxcourse.pitzuna.c
om. (64)
14:17:17.507663 IP (tos 0x0, ttl 122, id 24428, offset 0, flags [none], pro
to UDP (17), length 96)
    8.8.8.8.53 > 192.168.0.100.57202: 61419 1/0/1 linuxcourse.pitzuna.com.
A 84.248.108.122 (68)
14:17:19.463080 IP (tos 0x0, ttl 126, id 25461, offset 0, flags [none], pro
to UDP (17), length 80)
```

Image 2. DNS queries to 8.8.8.8.

The named site

The virtual host is in pitzuna.conf. ServerName is linuxcourse.pitzuna.com, the files are in /home/pitzuna/public-sites, and the host listens on port 3456. Require all granted lets visitors read that directory.

```
pitzuna@zunaserver:~$ cat /etc/apache2/sites-available/pitzuna.conf
<VirtualHost *:3456>
    ServerName linuxcourse.pitzuna.com
    DocumentRoot /home/pitzuna/public-sites
    <Directory /home/pitzuna/public-sites>
        Require all granted
    </Directory>
    ErrorLog ${APACHE_LOG_DIR}/pitzuna-error.log
    CustomLog ${APACHE_LOG_DIR}/pitzuna-access.log combined
</VirtualHost>
pitzuna@zunaserver:~$
```

Image 3. The virtual host.

I put the sentence "this is my public web server" in index.html. pitzuna.txt was written by my account and edituser.txt by edituser. curl with --resolve sent the name to 192.168.0.100 port 3456, and all three addresses returned their text. A file other than index.html is opened by adding its name after the slash.

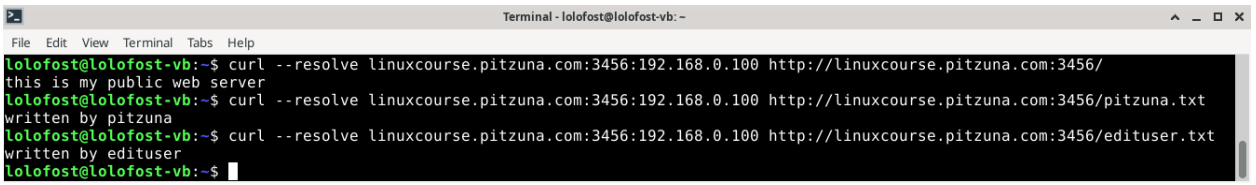


Image 4. The page and the two files.

Opening the name from inside the house would not have been a good test so I opened http://linuxcourse.pitzuna.com/ on my phone over the mobile network. The sentence came back, so the request had reached the server from outside.

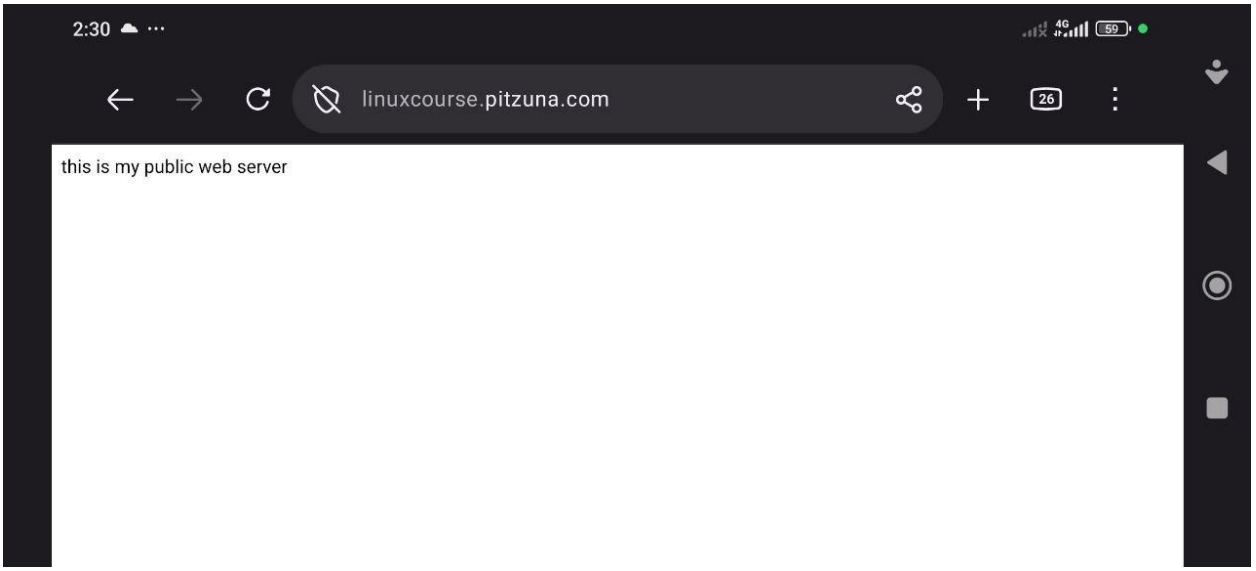


Image 5. The page from the mobile network.

Certificate

Certbot used the webroot in public-sites. The certificate was saved at /etc/letsencrypt/live/linuxcourse.pitzuna.com/ and expires on 7 January 2027.

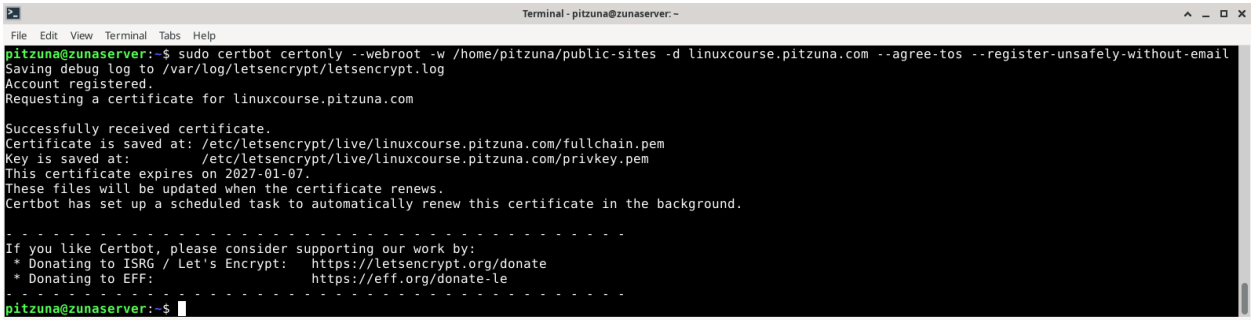


Image 6. The certificate was issued.

I decided to use Nginx Proxy Manager in the next steps as it is already set up on my server. I added the files created by certbot under SSL Certificates as a custom certificate in Nginx, selected that certificate on the proxy host, and turned on Force SSL.

https://linuxcourse.pitzuna.com/ then showed the sentence, and the shield icon said the certificate was verified by Let's Encrypt.

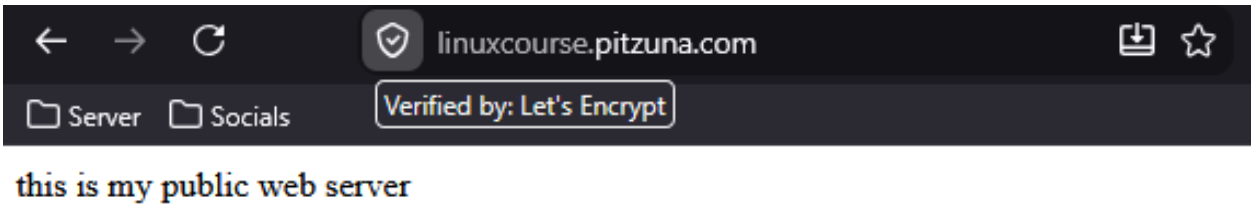


Image 7. HTTPS in the browser.

crt.sh returned 502 Bad Gateway, both for the front page and for the search. I've used an alternative: https://cc.la/certificates/linuxcourse.pitzuna.com It shows a valid Let's Encrypt certificate for linuxcourse.pitzuna.com, issued on 9 October 2026 and valid until 7 January 2027.

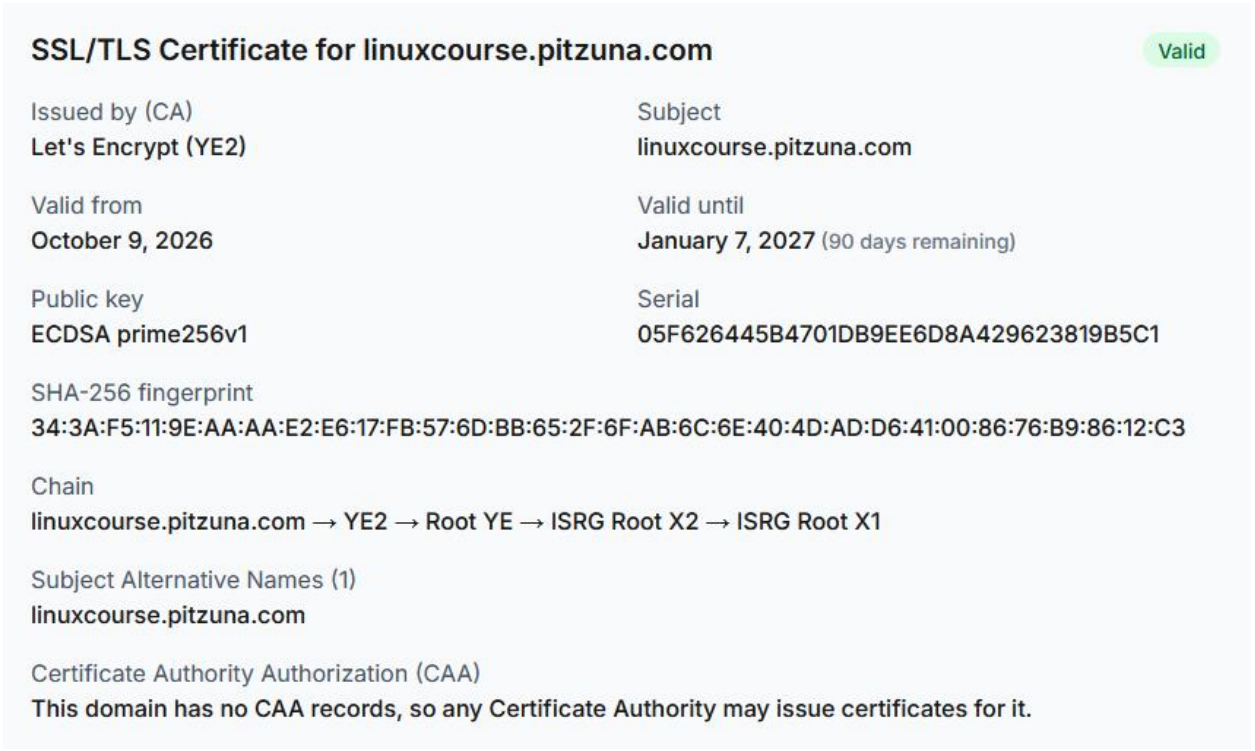


Image 8. The certificate on a public lookup page.

certbot renew --dry-run went through the renewal without replacing the certificate. It ended with the simulated renewal succeeded.

```
pitzuna@zunaserver:~$ sudo certbot renew --dry-run
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Processing /etc/letsencrypt/renewal/linuxcourse.pitzuna.com.conf
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Account registered.
Simulating renewal of an existing certificate for linuxcourse.pitzuna.com

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Congratulations, all simulated renewals succeeded:
  /etc/letsencrypt/live/linuxcourse.pitzuna.com/fullchain.pem (success)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
pitzuna@zunaserver:~$
```

Image 9. The renewal test.

HTTP and HTTPS

curl -v to port 443 completed a TLS 1.3 handshake. The certificate subject was linuxcourse.pitzuna.com, the issuer was Let's Encrypt, and the check said the name matched. The reply was HTTP 200 from openresty, and the page sentence was at the end. curl can print that sentence because it is the client and it decrypts the reply. A capture of the packets on the wire would not show the page.

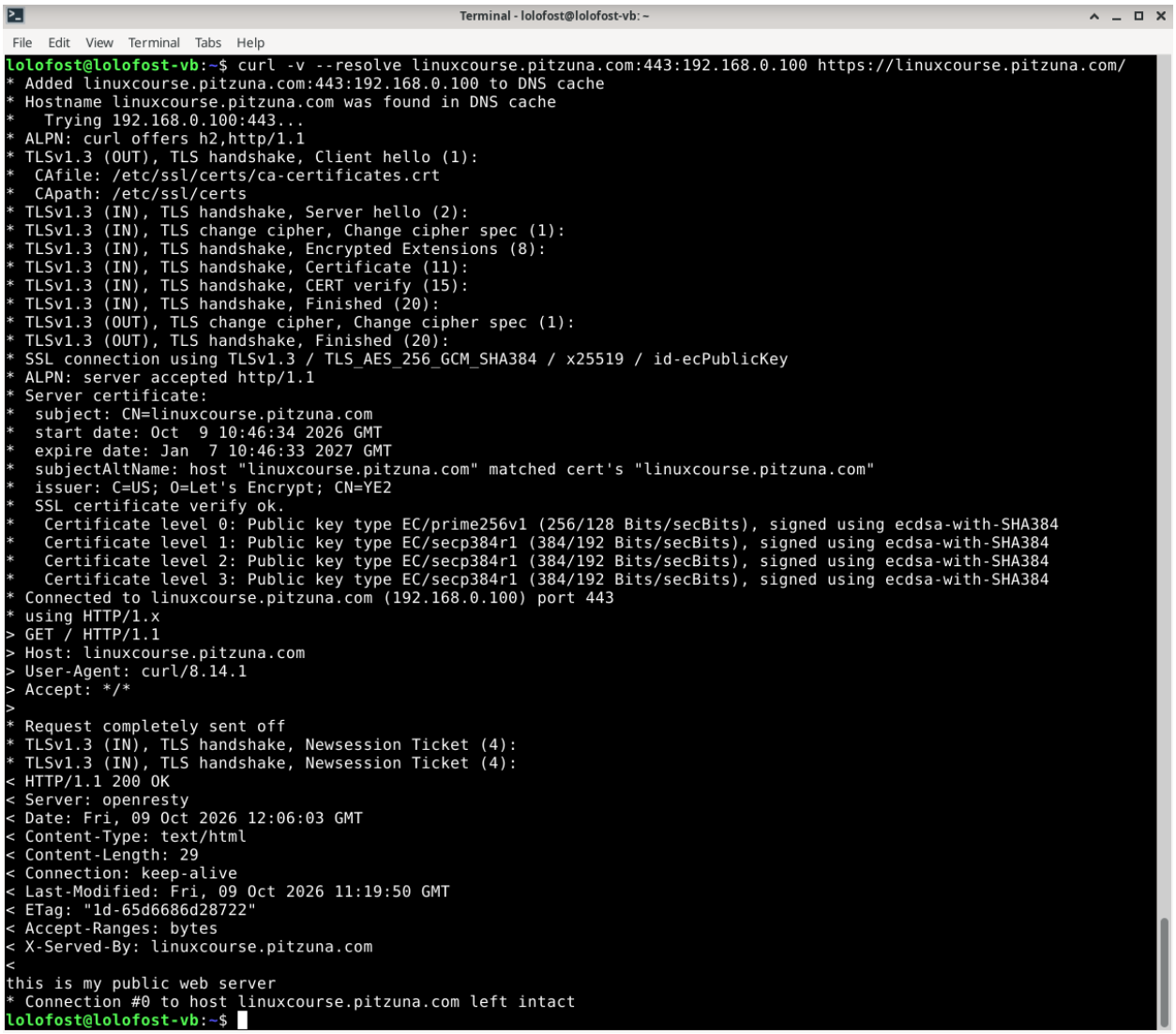


Image 10. HTTPS with curl.

The same command on port 80, with Force SSL still on, got HTTP 301 Moved Permanently. The Location header was https://linuxcourse.pitzuna.com/. The body was only the redirect page.

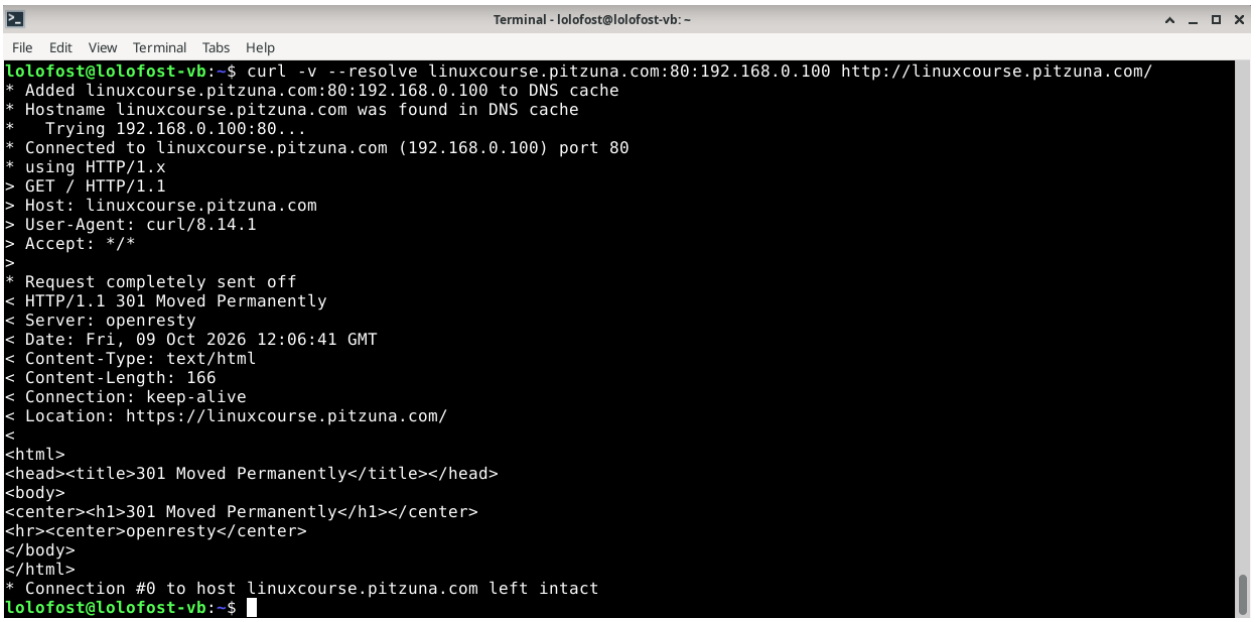


Image 11. HTTP redirects to HTTPS.

I turned Force SSL off and ran the HTTP command again. The status was 200 and the sentence was in the reply. I turned Force SSL back on after the picture.

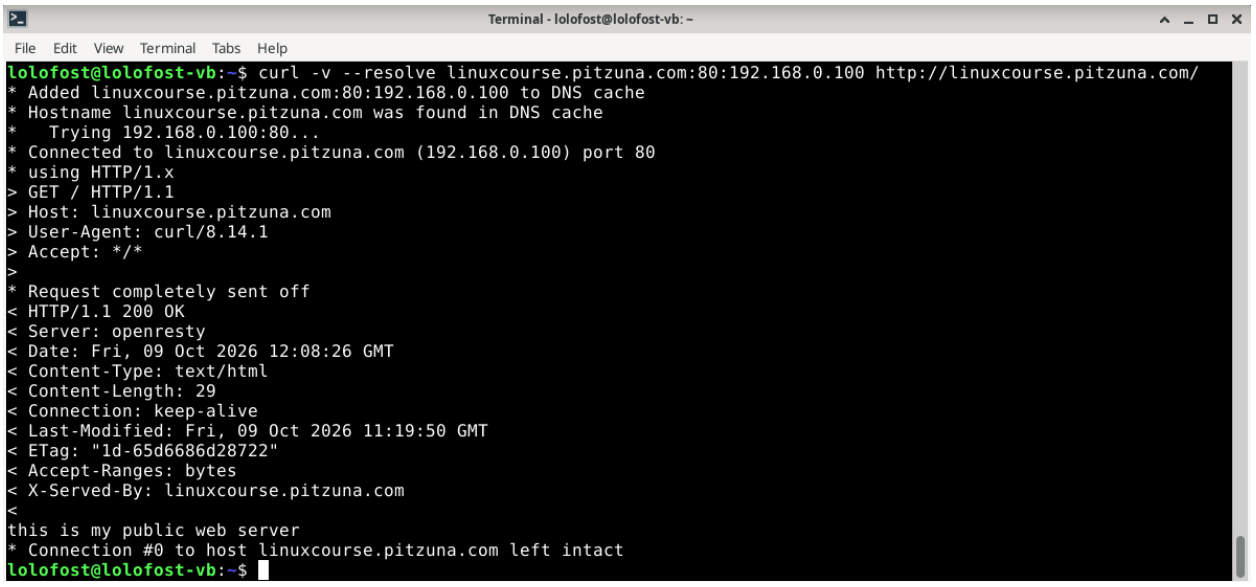


Image 12. HTTP while the redirect is off.